

CardioGuard AI

Real-Time Multi-Modal Cardiac Intelligence System

Comprehensive Technical Documentation & System Architecture

Digital Egypt Pioneers Initiative (DEPI) Graduation Project

Executive Summary

CardioGuard AI is an enterprise-grade Clinical Decision Support System (CDSS) engineered to monitor electrocardiogram (ECG) streams and detect life-threatening arrhythmias in real-time. Moving beyond traditional batch-processed machine learning, CardioGuard implements an event-driven big data pipeline capable of ingesting high-frequency physiological signals.

The core intelligence is powered by a hybrid deep learning architecture, replacing legacy BiLSTM models with a highly optimized Temporal Convolutional Network (TCN) intertwined with custom Attention layers and a 1D-CNN backbone. To ensure clinical trust and adoption, the system pioneers a dual-layer interpretability framework: visual spatial heatmaps via Explainable AI (XAI), and a Retrieval-Augmented Generation (RAG) pipeline that outputs bilingual, guideline-backed medical reports powered by Google Gemini.

Project Team & Engineering Domains

Team Member	Engineering Domain & Role
Yara Mohamed	Lead Systems & Knowledge Architect (Backend, Kafka, GenAI)
Jihad Ibrahim	Sequential AI Specialist (TCN & Attention Mechanisms)
Mohamed Emara	Signal Vision Expert (1D-CNN & Signal Preprocessing)
Osama Mahmoud	Azure MLOps Lead (CI/CD Pipelines & Deployment)
Faten	Data Synthesis & Digital Twin Lead (TimeSeriesGAN)
Menna ObyadAllah	XAI & Fusion Engineer (Grad-CAM, SHAP, Patient Metadata)

1. Systems Architecture & Data Streaming

Architected by Yara Mohamed

The foundational infrastructure of CardioGuard operates on an event-driven paradigm to decouple data ingestion from inference, ensuring zero data loss and horizontal scalability.

- **Apache Kafka Pipeline:** Raw 12-lead ECG signals are ingested at 500Hz via a containerized Kafka broker. Signals are logically partitioned by `patient_id` to guarantee strict temporal ordering. A continuous consumer daemon buffers incoming data into 10-second rolling windows (1000 timesteps) before triggering the inference graph.
- **Asynchronous API Gateway:** Built on FastAPI and Uvicorn, the REST API layer utilizes Pydantic for strict request validation. Heavy computational artifacts (Ten-

sortFlow weights and ChromaDB indices) are cached in the application state via asynchronous lifespan events to achieve sub-second endpoint latency.

- **Bilingual Medical Scribe (RAG):** Anomalous predictions automatically trigger a Retrieval-Augmented Generation pipeline. Utilizing BAAI/bge-small-en-v1.5 embeddings, the system queries official AHA/ESC guidelines from a local vector database, grounding the Google Gemini 2.5 Flash LLM to produce localized patient summaries in Egyptian Arabic alongside technical physician reports.

2. Hybrid Deep Learning Engine

Architected by Jihad Ibrahim & Mohamed Emara

The predictive core replaces standard recurrent networks with an advanced spatial-temporal topology, optimized for massive time-series sequences.

- **Spatial Feature Extraction (1D-CNN):** Engineered by the Signal Vision domain, a multi-layer 1D Convolutional Neural Network acts as the network's backbone. It efficiently filters high-frequency noise and baseline wander, extracting localized morphological features (e.g., QRS complex anomalies, ST-segment deviations) directly from the raw array.
- **Temporal Sequence Modeling (TCN & Attention):** To capture long-term cardiac rhythms without the vanishing gradient limitations of LSTMs, the Sequential AI domain implemented a Temporal Convolutional Network. Dilated causal convolutions establish a massive receptive field, while custom Attention mechanisms dynamically assign weight to critical cardiac events within the 10-second window, drastically improving multi-class arrhythmia detection on the PTB-XL dataset.

3. Explainable AI & Multi-Modal Fusion

Architected by Menna ObyadAllah

To transition from a "black-box" model to a trusted clinical tool, CardioGuard incorporates rigorous interpretability and individualized context.

- **Visual Interpretability:** Integration of SHAP (SHapley Additive exPlanations)

and Grad-CAM provides localized heatmaps over the raw ECG waveforms. This allows attending physicians to visually verify which specific cardiac cycle or electrical deviation triggered the anomaly alert.

- **Metadata Fusion:** Physiological waveform data is concatenated with deterministic patient metadata (Age, BMI, clinical history) in the fully connected layers, shifting the paradigm from generalized classification to individualized baseline modeling.

4. Data Synthesis & Digital Twin

Architected by Faten

Dataset imbalance presents a critical risk in medical AI, particularly regarding rare but fatal arrhythmias.

- **TimeSeriesGAN Augmentation:** Generative Adversarial Networks specifically engineered for time-series data were deployed to synthesize high-fidelity, mathematically sound ECG representations of minority classes. This ensures the TCN model generalizes effectively across the entire pathological spectrum without overfitting.
- **Digital Twin Monitoring:** By establishing a continuous baseline for a specific patient, the system performs ongoing drift analysis, serving as a rudimentary "Digital Twin" to detect gradual patient deterioration before an acute event occurs.

5. MLOps & Cloud Deployment

Architected by Osama Mahmoud

The entire ecosystem is designed for enterprise integration, adhering to strict MLOps principles.

- **CI/CD Infrastructure:** GitHub Actions govern the continuous integration pipeline, enforcing testing protocols before merging into the main branch.
- **Microsoft Azure Ecosystem:** Containerized artifacts are deployed via Azure App

Service, while model registry and versioning are handled securely through Azure Machine Learning, ensuring 99.9% uptime and seamless scalability to accommodate enterprise hospital workloads.

6. Key Performance Indicators (KPIs)

The platform's performance was evaluated against the following strict clinical and engineering benchmarks:

Metric	Target Benchmark
Streaming Ingestion Latency	< 200 milliseconds
Model Prediction Speed (Inference)	< 150 milliseconds
Multi-Class F1-Score (Arrhythmia)	> 95%
False Negative Rate (Critical Cases)	< 1.5%
API Uptime (Azure App Service)	99.9%